When Evidence Is Not Enough: Technical Supplement

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1 Scope and Evidence Status

This supplement accompanies the main paper and retains the broader research sequence, including weak baselines, unsuccessful interventions, diagnostic successes, failures to transfer, and resource interruptions. It is organized by evidential role rather than treating every development experiment as a confirmatory trial. Repeated questions, repeated controls and post-hoc diagnostic outputs are not pooled into a sample-size claim. No neural inference was run to prepare either manuscript. Historical files remain unchanged.

The paper package includes an artifact inventory with hashes, an unfiltered collection of 222 result/score/analysis documents selected by filename from the research and preserved-evidence trees, independently regenerated headline metrics, and raw-score checks. The collection is a machine-readable census, not 222 independent experiments. The complete research notes and `DECISIONS.md` remain authoritative for chronology. Superseded experiments, failed training fits and incomplete jobs remain in the repository and archives. The historical manuscript is preserved separately and should not be used for the new Ubuntu resource measurements.

Verification levels. All 20 handoff archives and their 2,930 mapped source files pass recorded byte-length and SHA-256 checks. The broader inventory records 3,776 research and preserved files, including duplicated evidence representations. Main Ubuntu results were recomputed from raw case ledgers: 1,920 initial/final programs were re-executed and rescored, all 960 records reconciled with the published export, and all stage-generated-token sums checked against the 2,048 ceiling. All 668,817 memory samples from 16 worker attempts pass the recorded RSS and process-swap checks. Fourteen Windows FinQA paired ledgers were re-executed and rescored using the pinned official evaluator and reported-answer scorer; seven TAT-QA prediction ledgers were rescored. The 96-context planner and 28-context secondary MAS outcomes were independently rescored using the official TAT-QA implementation. Older result documents are retained with their original audit provenance; the present checks do not claim independent expert adjudication of every financial label or a fresh replay of every historical training trajectory.

2 Reproducibility Specification

2.1 Study A: Frozen 16 GB Recipes

The Ubuntu run identifier is `lab-16-ubuntu-noswap-v1`. The frozen protocol identifier is `finance-deployment-lab-v2-115gb`. The public corpus contains 160 questions from 50 company-year reports (36 companies), selected from FinQA development data by report hash after recorded prior-exposure exclusions. This is not official-test performance. Six jobs combine two arms with primary, direct-control and repair-ablation profiles. The paired unit is the original question, with report clustering; 960 outputs do not represent 960 independent questions.

Setting	Recorded configuration
Single model	Qwen2.5-14B-Instruct, GGUF Q4_K_M; no project adapter.
Resident team	Qwen2.5-1.5B-Instruct F16 source suggestion; 3B F16 program drafting; 0.5B F16 binary verification. Separate simultaneously loaded model instances, no shared backbone.
Inference	CPU only, 8 threads, context 4,096, greedy temperature zero, seed 42, GPU layers zero, `use_mmap=False`.

Setting	Recorded configuration
Output budget	2,048 generated tokens per case across all stages. Actual token use differs by arm/profile; prompt tokens sum every call, not one context window.
Primary	Analysis allowance 128; two fixed training demonstrations; binary review; no repair.
Direct control	Removes both analysis and demonstrations, retaining the structured program and review pipeline. It is not the direct-number Windows comparator.
Repair ablation	Same primary initial trajectory plus full regeneration/review after failure. Initial trajectories are matched; later candidates may change.
Evidence	Lexical IDF/length-normalized top-eight source units; compact quantity catalog; answer keys absent from generation packets. MAS receives source suggestions.
Runtime	`llama-cpp-python` 0.3.16, NumPy 2.5.4, psutil 7.2.2; Python 3.12.15; Linux 7.0.0-38-generic, glibc 2.43, x86-64.
Hardware	AMD Ryzen 5 8645HS; Linux physical RAM recorded as 15,438,376,960 bytes. GPU is not used for these inference runs.
Resource control	11,500,000,000-byte sampled controller-plus-worker RSS guard, process-swap monitor, and user scope reporting `MemorySwapMax=0`; not a hard RSS cgroup limit.
Resume	Original manifests and per-attempt traces retained across interruption/reboot. A one-page usable-RAM reporting drift was reconciled only with the remaining guarded identity fields matching.

Model-file and source SHA-256 values are preserved in `reproducibility/identities.json`; no large model tensors are bundled with the paper. The original source public-packet hash is `9611c2f650983b9cd8d04823824cbf1999d6bbb00fea9cea3c9894875cb9bf59`. Exact prompt text and responses for every original stage are in the restored case ledgers. Representative complete prompts, selected mechanically as the first recorded case in each job, are included in `reproducibility/prompts/`. They are examples of recorded inputs, not replacements for the frozen templates or full case-specific evidence.

2.2 Windows 8 GB Application Budget

Windows 11 build 10.0.26200 runs on an AMD Ryzen 5 8645HS with 6 physical/12 logical CPUs and 16,390,729,728 physical bytes. The inference executable is `llama.cpp` b12202, revision `d6be06557`, x64 CPU. Its distribution archive hash is retained with the environment record. The reported environment includes Python 3.12.14 and psutil 7.2.2. Both resident workers use context 4,096 and eight CPU threads; execution of drafting/review stages is sequential while both weights remain loaded. The shared Job Object has a 6,100,000,000-byte commit limit, and the separate working-set guard is 6,400,000,000 bytes. The initial Windows pilots used sampled RSS only; tables and narratives do not upgrade their guard retrospectively.

The direct/program 7B comparator uses the two-shard Qwen2.5-7B-Instruct Q4_K_M checkpoint. Small-model conditions use the recorded Qwen2.5-3B-Instruct Q4 checkpoints; the matched Q4/Q8 study quantizes one shared F16 source and must not be conflated with comparisons using other pre-existing files. Role-adapted runs attach the common actor or control critic LoRA to the appropriate separate 3B instance. Artifact names, sizes and hashes distinguish these identities in the archived model and launch records. All paper generation is based on those identities; matching a parameter count alone is not sufficient to reproduce a condition.

For the held-out planner, temperature is zero and the common generation ceiling is 960. Current MAS reserves 768/192 for draft/review; planner MAS reserves 192/576/192 for shape/binding/review. A rejected shape still consumes its planning work and falls back within the 576-token drafting allowance. Direct/program 7B each reserve 960 without review. Their sequential CLI duration includes loading; resident pair request times exclude initial worker loading. The measured latency difference between the two resident arms is the cleanest planner contrast. Prompt RAM caching is disabled with `--cache-ram0`; context length is unchanged.

2.3 Adapter Provenance and Training

Adaptation is an established baseline component (Hu et al., 2021; Dettmers et al., 2023), not evidence of algorithmic novelty. Initial 3B adaptation uses an NF4 training base from `unsloth/Qwen2.5-3B-Instruct-bnb-4bit`, pinned revision `2672b588848527e456d320c11926c794539f47d5`. The deployment GGUF Q4_K_M base is a different quantization; exact full-precision weight equivalence is not asserted. Invalid source-binding corpus revision 01 was superseded and is not presented as usable training data.

The valid initial corpus has 761 examples fitting the full 1,024-token context constraint, split 661 train/100 validation by recorded report groups. The completed initial epoch uses 83 optimizer updates and 14,966,784 trainable parameters across 504 changed adapter tensors. A full-logit fit failed with GPU OOM before its first update. The successful supervised-suffix-logit implementation retains all context and supervised targets; software equivalence checks and both failed/successful attempts are retained. Teacher-forced loss is not QA accuracy. Training has its own GPU/host memory accounting and is not claimed to fit the CPU inference tier.

The common actor continues from the initial actor with 128 training examples, 32 validation examples and 16 optimizer updates. The matched control critic continues from its recorded initializer with 384 training examples and 48 updates. Its validation groups contain 192 crossed, 192 mutation and 32 replay-critic examples. The continuation recipe uses learning rate 10^{-4}, zero weight decay, clipping norm 1, accumulation 8, one epoch, seed 42, optimizer reset, rank 8, LoRA alpha 16, dropout 0.05 and all-linear target modules. Training versions include PyTorch 2.6.0+cu124, Transformers 4.57.1, PEFT 0.17.1, bitsandbytes 0.48.1 and Accelerate 1.10.1. Initializer, data, optimizer and conversion manifests are retained. Native application checks establish that adapters alter outputs, not that they improve QA. The converted common actor SHA-256 is `4e21beca69dcf7bf2b9e89df7d189c74740d98acebe08501fd7e848bb7155d3e`.

2.4 Cohorts, Exposure and Prompt Availability

FinQA train/development/test snapshots share no exact question or page IDs in the recorded audit but do overlap at company-year level: 243 of 278 test reports appear in the raw training source and 93 in raw development. Exclusion of actually used fitting/validation reports differs from exclusion of an entire public source split. We describe exclusions per study, not as a universal "unseen test" guarantee. The 96-report precision/catalog/format cohort is shared, not three independent replications. The oracle matrix uses the same 15 enriched reports in every arm. The 72-context TAT-QA format comparison reuses a hash-ordered subset of the 120-context capacity cohort; the separate 72-context adaptation-transfer cohort excludes earlier contexts. The held-out planner primary and secondary cohorts exclude prior table IDs and exact table duplicates, but no report mapping or pretraining audit is available.

The TAT-QA dataset/evaluator source is pinned at repository revision `870accc41953dcde885aabeb963d94aabdc0fbc3`. Downloaded evaluator and data bytes match all recorded hashes. Primary/secondary planner packet hashes are respectively `e463671e33f56af56e6d8d1ad07720b3e75fa3c9a732746a541483fb44251ac0` and `f494a4268baa1623b8159dc2d428c0cfaeb3235e99df5e9831bf923dcaa1a802`, using LF line endings. TAT-QA is attributed to its authors (Zhu et al., 2021); the data card specifies CC BY 4.0. Third-party source files retain upstream provenance; inclusion does not change their licenses.

Windows resident ledgers retain responses, usage and decisions but do not embed every rendered prompt. Exact prompts are reconstructible from the frozen packet and versioned pipeline/template code. The paper package records these source hashes and includes the relevant template files; it does not falsely describe missing rendered prompts as observed logs. Template text includes question/evidence, source catalog, program grammar and scale instructions; review sees the candidate and deterministic execution without labels. The planner uses only operations, dependencies and VALUE placeholders. Complete reference programs appear only in explicitly labeled oracle diagnostics and controlled candidate challenges.

3 Metrics and Statistical Details

Historical reported-answer score. The evaluator parses supported simple numerical reported labels and their rounding precision. Predicted percentage points are divided by 100; percent labels require percent/fraction output units, while ordinary numerical labels require number/fraction units. The normalized value must fall within the larger of reported rounding tolerance and 10^{-5}, with a valid ANSWER decision and no execution issues. Unsupported

labels remain in the denominator. This historical rule can disagree with the raw execution reference and should not be interpreted as official program equivalence.

Raw execution score. For finite predicted $\hat{y}$ and raw reference y, the numerical condition is $|\hat{y} - y| \leq \max(10^{-5}, 10^{-5}|y|)$. Correct delivery additionally requires the pipeline to deliver. The stored raw-correct field is a numerical score; source support and review are separate. Primary SAS has 34 deliveries correct by both scores, 44 reported-only, four raw-only and 44 wrong by both. Thus 48/126 delivered answers change correctness label; the two scores are not nested. Primary MAS has five correct by both and three wrong by both.

Official benchmark evaluators. Windows FinQA program outputs are translated to the benchmark's arithmetic token representation and evaluated using the pinned implementation, then compared with stored execution labels. Its evaluator hash is `845cd131cab843eceff256cf6d392978cc470a7da4a80107beb56027fdca5c13`. TAT-QA uses the authors' `TaTQAEmAndF1` implementation, hash `2aeeac479f89f8c76300af1cc0e8d098eb86af84bc386b38b6ab4af484a6dea8`, with predicted numerical scale. On the reported arithmetic single-number comparisons, summed F1 equals summed exact match; it is not an additional independent success endpoint.

Cluster bootstrap. For Study A, let cluster g contain n_g questions and paired difference sum s_g. Each draw samples the 50 reports with replacement and computes $\sum_g s_g / \sum_g n_g$ over sampled clusters. We use 10,000 draws, NumPy `default_rng(42)`, and linear 2.5/97.5 percentiles. The primary reported correct-delivery interval is $[-0.538961, -0.365517]$ and company-cluster sensitivity is $[-0.534722, -0.364405]$. The corresponding raw interval is $[-0.282609, -0.133720]$. These are resampling estimates under the observed cluster structure, not randomized-treatment causal intervals. The original sealed analysis uses Python RNG/quantile indexing; its original interval $[-0.540373, -0.366906]$ is retained, not overwritten.

Planner paired intervals reproduce the archived Python `random.Random(73)` procedure: 10,000 draws of 96 table-context differences, sorted indices 250 and 9749. For gains b, losses c, the two-sided exact McNemar value is $\min[1, 2\sum_{k=0}^{\min(b,c)} \binom{b+c}{k} / 2^{b+c}]$. With four gains and zero losses it is 0.125. Empirical resampling cannot invent a loss absent from four observed discordances, explaining why the positive percentile interval must not be presented as strong confirmatory evidence. The zero-width secondary correct-delivery interval is a degenerate empirical bootstrap, not proof of population equality.

Older Windows paired summaries generally use 2,000 company-cluster draws with Python seed 42 and indices 50/1949; we retain that procedure instead of silently substituting a different interval. Exact question-pair tests assume independence of pairs; shared companies may violate it. Older 16 GB supplemental analyses include company sensitivity, equal-report weighting, leave-one-report-out checks and cluster sign-flip tests; these were post-hoc, with sign-exchangeability assumptions, and are retained in the original statistical package. The new paper does not elevate them into prespecified primary discoveries. No observed-power, noninferiority or equivalence claim is made. No seed/model uncertainty is inferred from a question bootstrap.

4 Complete Main-Study Outcome and Resource Tables

	Reported			Raw			Hours	RSS GB
Profile / arm	C	W	H	C	W	H		
Primary SAS	78	48	34	38	88	34	11.668	10.075
Primary MAS	5	3	152	5	3	152	3.383	10.946
Direct SAS	50	66	44	14	102	44	5.314	10.070
Direct MAS	0	5	155	0	5	155	0.897	10.934
Repair SAS	79	50	31	38	91	31	13.598	10.087
Repair MAS	5	3	152	5	3	152	3.781	10.946

Table 2: All Ubuntu conditions, $n = 160$ each. Hours sum saved inference-stage time, excluding startup, interrupted work and restart overhead. Memory is sampled process-tree RSS in decimal GB.

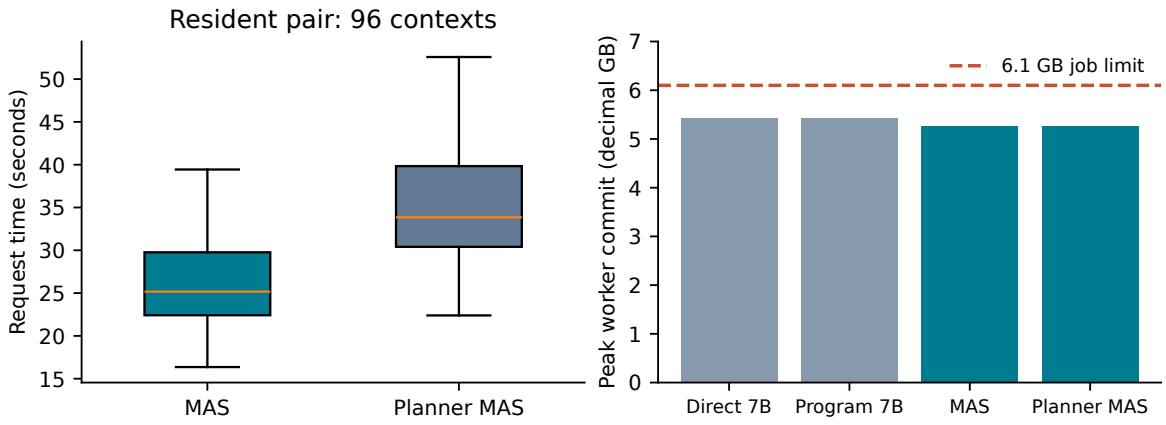


Figure 1: Primary held-out planner resources. Left: distribution of resident request time; boxes show interquartile ranges, median and standard 1.5-IQR whiskers, with outliers omitted from display but retained in data. Right: measured peak worker commit, not physical RAM or an uncertainty interval. Direct/program 7B timing includes load and is therefore reported separately in text.

The primary/repair initial programs, scores, execution and verdicts are matched. The analysis stage intentionally reaches its 128-token allowance on 159 SAS and 157 MAS primary questions; these events are not malformed final JSON or context skips. The sealed analysis reports zero context/budget-skipped calls in the six jobs. Removing unused operations by backward slicing preserves executable numerical values; selecting an earlier endpoint is a different, potentially answer-changing intervention.

Cohort	Arm	C	W	H	Correct candidates
Primary 96	Direct 7B	19	76	1	19
	Program 7B	15	80	1	15
	Current MAS	6	23	67	26
	Planner MAS	10	25	61	25
Secondary 28	Current MAS	2	10	16	5
	Planner MAS	2	10	16	3

Table 3: Held-out TAT-QA official exact-match outcomes. Direct/program candidates equal final predictions because those arms have no review. Secondary 7B jobs are incomplete and are not assigned whole-cohort scores.

Primary baseline outcome	Planner correct	Planner wrong	Planner withheld
Correct	6	0	0
Wrong	2	19	2
Withheld	2	6	59

Table 4: All 96 paired delivery transitions. Four correct gains coexist with six newly wrong deliveries and four removals of wrong delivery (two corrected, two withheld).

Secondary transitions are two correct-to-correct, nine wrong-to-wrong, 15 withheld-to-withheld, one wrong-to-withheld and one withheld-to-wrong. Thus equal aggregate wrong counts do not mean identical behavior. Both correct delivery identities are unchanged. Seventeen of 28 predicted shapes are usable.

5 Exploratory Sequence: Positive, Null and Adverse Findings

5.1 Earlier Linux Development and Training

The original structured synthetic development suite is solved by a generator-aware deterministic parser (150/150); the 24 temporal KYC probes are also solved by a compact rule control. These are regression/mechanism tests and cannot establish the necessity or superiority of an LLM. A four-case shared-weight pilot improves delivered coverage from zero to one while candidate correctness remains one; patch repair has no gain. Shared residency and successful fitting establish resource behavior, not semantic capability.

On the repeatedly reused first 24 FinQA development questions, the stronger 7B SAS with 128-token analysis and two demonstrations delivers 13 correct, five wrong and withholds six by reported scoring. Longer 256-token analysis yields 11 correct and nine wrong; more table coverage leaves 13 correct. Symbolic cancellation feedback corrects no answers beyond plain retry in a two-trigger pilot. The 3B MAS with analysis/demonstrations produces four reported-correct candidates but delivers three correct, one wrong and withholds 20. Longer analysis does not improve its four correct candidates and increases wrong delivery to ten. These are development comparisons, not optimized-system bounds.

Matched no-analysis/no-demonstration actor adaptation changes 0/24 reported-correct candidates (all refusals) to 11/24, with ten correct, 11 wrong and three withheld. The gain is real on reused data, but starts from a weak floor baseline. A larger unadapted 3B verifier with the same 11 correct drafts withholds all 24, demonstrating that stronger nominal verifier capacity alone does not ensure useful selective delivery.

Matched critic continuation on 52 questions/15 report groups yields control 3/19/30 and crossed 3/15/34 by reported scoring; the wrong-rate interval includes zero. Against the stronger original critic's 4/22/26, crossed loses one correct delivery and fails the no-correct-loss rule. Repair changes no final outcomes relative to saved no-repair prefixes. A reserved 49-question/15-report diagnostic yields control 4/17/28 versus crossed 4/9/36 under reported scoring, but raw correct delivery falls from eight to seven. Selecting only the favorable metric would conceal failure of the both-metric adoption criterion. SAS and additional baseline conditions are all retained in the condition census below.

The matched SAS continuation on the same 52-question selection cohort changes reported outcomes from 9/20/23 (control) to 9/10/33 (crossed), but raw correct delivery decreases from 11 to ten. The unadapted analysis/demonstration baseline gives 9/24/19 reported and 10/23/19 raw. Thus the crossed SAS can improve wrong delivery against that baseline without an observed correct-count loss, yet still fails the both-metric no-loss comparison against the matched trained control. These are different reference systems, not interchangeable success claims.

On a controlled 12-report/24-program mutation probe, a trained critic accepts 11 of 12 reference programs and zero mutations; the plain reviewer withholds all. A question-blind depth-three program-feature tree scores 275/382 on original synthetic validation, and an extended-corpus tree 295/382 on its own validation; both exceed the 50% majority baseline without seeing questions. On the reused natural24 pool, original/extended trees yield reported 5/8/11 and 5/5/14, versus the original neural critic's 11/4/9. These positive neural and shallow-control results show both discriminative potential and shortcut opportunities, not proof of neural shortcut use or broad semantic understanding. The neural gain remains development evidence.

A source-header materialization prototype adds 183 catalog quantities while preserving all 1,922 original quantities across 73 fitting packets, and restores reference operand coverage on four questions. A proposed 292-case-arm neural comparison was prepared but not run. Executable source support is verified; a QA gain is not measured. It is not counted as a completed method success.

External exploratory controls use original unadapted recipes: MortarBench local typed matching is 39/90 for SAS and 26/90 for MAS, based on partial JSON/XML evidence from nine prepared loan cases; this is not the official full benchmark protocol. The FRAMES closed-book control gives 2/100 and 1/100 under strict matching, unchanged by punctuation sensitivity; evidence URLs are not retrieved. These controls do not establish retrieval or multi-hop generalization. Their scoring and exposure limitations prevent treating them as confirmatory evidence for the adapted financial program systems.

5.2 Windows Candidate and Verifier Pilots

The initial planned 24-pair Windows candidate comparison stops after five complete pairs for post-hoc diagnosis. Refusal-allowed yields five refusals. Forced attempts yield zero reported-correct and one raw-correct candidate; a forced-one-step follow-up on the same five yields two reported-correct and one raw-correct. The original reviewer

withholds everything. Removing contradictory repair fields or using a verdict-only 0.5B reviewer accepts all five (two reported-correct, three wrong); a 3B verdict-only reviewer withholds all. This exposes a local schema-policy problem but not semantic discrimination.

On four new reports with reference/near-miss candidates, the 0.5B reviewer accepts all eight and the 3B reviewer withholds all eight. Forced 3B comparison selects the reference in five of eight order presentations, only two of four reports in both orders. On six new reports with a reject-both option, a reversed-order agreement rule yields three correct, zero wrong and nine withheld across the two pair types per report. These dependent, reference-enriched challenges do not estimate natural-pool delivery risk.

The next six-report natural-pair pilot produces no correct final candidate in either draft and withholds all six. A separate six-report base/adapted pilot produces zero versus one correct candidate, yet agreement delivers zero correct, two wrong and withholds four. Thus the apparently error-free constructed result fails to transfer. A six-report adapted-3B versus unadapted-7B program pilot produces one versus zero correct final candidates, with more resources used by 7B; an answer-key prefix selector could recover four reports, but no deployable endpoint selector is demonstrated.

5.3 Endpoint, Retrieval, Precision and Catalog Controls

Model-predicted operation count on 16 reports changes official correct candidates from one to three (three gains, one loss). A gold-count control also yields three, on exactly the same recovered reports; gold count adds no recovery beyond predicted count on that slice. A retrospective endpoint selector makes no change in its initial test. On the 96-report matched precision cohort, a first-step rule changes Q4 official accuracy from 2 to 8 and Q8 from 2 to 6. On the earlier 24 reports it instead reduces Q4 from 3 to 1 and Q8 from 5 to 1. This cross-cohort reversal prevents a general recommendation to truncate at the first operation. The answer-key prefix oracle reaches at most 11/96 per arm, leaving most errors outside prefix selection.

Unused operations occur in 67/96 Q4 versus 14/96 Q8 outputs (and 20/24 versus 7/24 in the earlier cohort). Slicing preserves final execution results; it does not create an accuracy gain. Precision raises initial official matching from 3/24 to 5/24 (one loss, three gains; $p = 0.625$), with unchanged 3/24 normalized symbolic matches. Its 96-report replication is 2 versus 2; reported scoring also ties but with one discordance in each direction rather than the two official discordances each way. Q8 raises peak commit from about 2.59 to 3.94 GB and median latency from 20.6 to 32.2 seconds in that replication.

Retrieval screens on 883 development questions cover all annotated spans for lexical8 750, BM25 748, reciprocal-rank fusion 733, BGE-small 728 and BGE/lexical hybrid 748. Wider lexical retrieval covers 827, adding 77 with no losses, at a larger token allowance (median evidence/catalog tokens 711 to 1,225). Numerical reference bindability rises 703 to 767; source identity and model success remain separate. The pilot's 1/16 to 4/16 generation gain ($p = 0.25$) fails directionally at 2/21 to 1/21 ($p = 1$). The latter company-cluster interval is $[-20.8, +11.1]$ percentage points, and median latency rises 18.9 to 26.3 seconds. A fixed-catalog follow-up was proposed to separate additional text from catalog options, but no completed result is identified in the retained retrieval evidence; it is not counted as an executed ablation.

Catalog completion makes 28 additional annotated test programs numerically bindable in its source screen. Five enriched neural cases improve 0 to 2; on the unselected shared 96 reports accuracy changes 2 to 1, despite reference-executable witnesses increasing 82 to 84. On the separate 15-report oracle matrix, repeated unrestricted controls reproduce their plans. All four structure-only successes remain correct with the shortlist. Structure plus shortlist yields ten official and seven reported successes; it fixes multiple variables simultaneously and is neither a deployable method nor a single-factor mechanism attribution.

5.4 Baseline Capacity, Format, Review and Reference Checks

The initial 16-context TAT-QA format pilot gives 1/16 for both 3B direct and program output, on different questions. The 40-context direct-capacity comparison gives 5/40 for 3B and 6/40 for 7B (four 3B-only, five 7B-only); a 120-context replication gives 12 versus 29 (five versus 22 unique successes). Gold-scale substitution rescues 11 wrong 3B and five wrong 7B answers in that replication, showing that scale selection matters but does not explain the full gap. This is an answer-key sensitivity, not deployed accuracy.

Fixed-7B format controls favor direct output on FinQA's shared reported metric, 35/96 versus 20/96, but favor programs with substantial uncertainty on TAT-QA, 19/72 versus 14/72. The FinQA agreement gate retains only three correct, two wrong and withholds 91, losing 17 of 20 reported-correct programs. Its percentage-label subgroup is direct

23/53 versus program 19/53; other supported numerical labels are 12/40 versus 1/40; three unsupported labels receive no match. The original official program score 1/96, converted sensitivity 19/96 and reported score 20/96 are retained separately.

The initial trained-critic replay on 16 wider-evidence candidates appears favorable under reported scoring: control keeps three correct with zero wrong. The transfer replay on 21 narrow candidates has an executable-only baseline of two correct/19 wrong; control yields 1/5/15, crossed 1/2/18. Both lose one correct candidate, and two of three crossed-accepted outputs are wrong. Independent 7B recomputation on the initial 16 yields four correct answers alone, but the agreement gate withholds all; one overlapping correct answer pair differs beyond the frozen tolerance despite matching the reported label separately.

On 72 separate TAT-QA contexts, actor adaptation improves candidates from 3 to 8, and review delivers 6/10/56; ten of 16 accepted answers are wrong. A resident 21-report workload produces six official-correct (five reported-correct) candidates and delivers 5/4/12 under both scores. Disabling RAM prompt caching preserves all plans/verdicts/deliveries, reduces sampled RSS from 5.852 to 5.013 GB and job peak from 6.101 to 5.264 GB. The default-cache job logs failed allocation and fallback, whereas the disabled-cache run logs none. A failed-allocation peak does not identify exactly how many bytes were successfully committed.

A temporal growth rule applied to 453 questions from 117 report-disjoint test reports delivers only three, none matching either benchmark score. Individual discrepancies include plausible financial interpretations and incompatible reference conventions; they are not adjudicated gold-label corrections. A broader consistency screen reproduces every stored annotated execution result for 883 development and 1,147 test questions, then flags reported/raw incompatibility on 110/851 eligible development labels and 129/1,100 eligible test labels. These are screening flags, not 239 proven annotation errors. Official primary labels and all cases are retained.

5.5 Planner Development and Held-Out Result

The initial 12-report prototype produces a usable shape on only one report; 11 predictions are constant-only. Revising the grammar to unbound VALUE slots yields usable shapes on 11/12. Official final correct deliveries remain three in both arms; wrong decreases four to three and withheld increases five to six. Reported-correct delivery falls five to four. Median latency rises 23.0 to 30.9 seconds. This is development evidence for producing valid shapes, not an accuracy gain. The revision precedes held-out inference.

In the main 96-context cohort the four correct-delivery gains have no losses, but wrong delivery rises two and candidate correctness falls one. The main paper reports this failed practical criterion prominently. At 28 secondary contexts there is no correct-delivery change, and two correct candidates are lost. The incomplete 7B test prefix and recovery attempt remain archived; completion of MAS does not imply completion of the four-arm comparison. No new test experiments are substituted for those failures.

6 Resource Failures, Missing Evidence and Scope of Claims

Early Linux training OOMs, converter/application interface issues, capacity probes, historical process swap, interrupted external evaluation, default-cache allocation fallback, secondary 7B allocation failures and Ubuntu SSD disconnections are retained. They are neither silently excluded nor pooled into successful end-to-end timing. The original 16 GB recovery run's score completeness did not validate its no-swap claim. The Ubuntu run supplies a separate resource-qualified repeat, with the same questions and scores.

The SSD was mounted read-only after NTFS metadata errors. Six unreadable replay JSONLs were reconstructed using sealed analysis code and pinned 0.3.16 vocabulary-only tokenizers; their hashes, all 40 analysis-evidence hashes and the entire summary match the original audit. The text report was regenerated without an original recorded hash. The original results tarball and one bootstrap log remain unavailable, and a workflow marker still reads RUNNING. Therefore evaluation and replay completion are supported, but successful original packaging is not. Repository handoff ZIPs verify independently. Model tensors were not rehashed during manuscript preparation and are not included in portable evidence.

Missing or limited evidence includes physical 8 GB/no-paging validation, an uninterrupted full Ubuntu run, complete secondary 7B baselines, TAT-QA report-level grouping, multiple model families/seeds, independent financial annotation adjudication, optimized baselines for every method and matched-latency system initialization. None is filled

with estimated or fabricated data. No measured 32 GB result exists. The paper makes no calibrated safety, financial compliance, causal architecture or universally optimal precision claim.

7 Literature Verification and Reproduction Commands

The bibliography contains 17 references verified against primary arXiv metadata, ACL Anthology or PMLR publisher BibTeX. The accompanying literature audit records the claim supported by each source and comparison limits. Recent financial verification papers are identified as arXiv preprints; we do not invent venues or reproduce their performance under our budgets. Our numerical results come only from this repository, not from unrelated published leaderboards. LoRA, QLoRA, program execution, selective prediction and multiagent debate are prior methods, not originality claims.

From a repository checkout, restore immutable evidence into a separate directory, substituting its path for PATH:

```
python dashboard_handoff/restore_evidence.py --extract PATH
python scripts/reporting/build_final_paper.py --evidence PATH
python scripts/reporting/verify_final_scores.py --evidence PATH
```

These commands use preserved predictions and do not load model weights or run inference. Dependencies and exact input hashes are recorded with the paper. Compile each LaTeX source using PDFLaTeX, BibTeX and two additional PDFLaTeX passes. The main paper uses unmodified AAAI-2027 style/BibTeX files; this separate technical supplement uses one column for audit readability. Submission-specific administrative declarations, final author identities and any required checklist require author review; none is fabricated.

8 Historical Linux 8 GB Condition Census

The following is generated directly from the preserved condition inventory, without selecting conditions by outcome. It covers the exported natural-task development/selection/diagnostic conditions. Separate Windows experiments, synthetic regression tasks, external controls and unsuccessful execution attempts are described above and retained in the machine-readable artifact inventory. The labels retain original run names to make each row findable. R and X are the disclosed reported and raw historical scores, not interchangeable with official FinQA equivalence.

Table 5: Historical Linux 8 GB condition census. R: reported; X: raw execution. C/W/H retain all cases. These overlapping development, selection and diagnostic conditions are not independent replications.

Condition / metric	n	C	W	H
diagnostic_audit-control_mas-full-01 / R	49	4	17	28
diagnostic_audit-control_mas-full-01 / X	49	8	13	28
diagnostic_audit-control_sas-full-01 / R	49	8	20	21
diagnostic_audit-control_sas-full-01 / X	49	14	14	21
diagnostic_audit-crossed_mas-full-01 / R	49	4	9	36
diagnostic_audit-crossed_mas-full-01 / X	49	7	6	36
diagnostic_audit-crossed_sas-full-01 / R	49	9	20	20
diagnostic_audit-crossed_sas-full-01 / X	49	15	14	20
model_selection-baseline_mas-full-01 / R	52	2	19	31
model_selection-baseline_mas-full-01 / X	52	3	18	31
model_selection-baseline_original_classic_mas-full-01 / R	52	4	18	30
model_selection-baseline_original_classic_mas-full-01 / X	52	6	16	30
model_selection-baseline_original_critic_common_actor-full-01 / R	52	4	22	26
model_selection-baseline_original_critic_common_actor-full-01 / X	52	6	20	26
model_selection-baseline_sas-analysis128-demos2-full-01 / R	52	9	24	19
model_selection-baseline_sas-analysis128-demos2-full-01 / X	52	10	23	19

Condition / metric	n	C	W	H
model_selection-baseline_sas-full-01 / R	52	8	25	19
model_selection-baseline_sas-full-01 / X	52	10	23	19
model_selection-control_mas-full-01 / R	52	3	19	30
model_selection-control_mas-full-01 / X	52	5	17	30
model_selection-control_sas-analysis128-demos2-full-01 / R	52	9	20	23
model_selection-control_sas-analysis128-demos2-full-01 / X	52	11	18	23
model_selection-control_sas-full-01 / R	52	8	23	21
model_selection-control_sas-full-01 / X	52	10	21	21
model_selection-crossed_mas-full-01 / R	52	3	15	34
model_selection-crossed_mas-full-01 / X	52	5	13	34
model_selection-crossed_sas-analysis128-demos2-full-01 / R	52	9	10	33
model_selection-crossed_sas-analysis128-demos2-full-01 / X	52	10	9	33
model_selection-crossed_sas-full-01 / R	52	10	16	26
model_selection-crossed_sas-full-01 / X	52	12	14	26

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